CS5487 – Assignment 3 – Course Project

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Proposal due date: Fri, Week 10
Presentation date: Fri, Week 14
Report due date: Sun, Week 14

1 Course Project

The final assignment is a student-defined course project. The goal of the project is to get some hands-on experience using the course material on your own research problems. If you can't think of a project, then you can do the "default" project, which is digit classification (see Section 2).

1.1 Project topic

The goal of the project is to get some hands-on experience using the course material on your own research problems. Keep in mind that there will only be about 4 weeks to do the project, so the scope should not be too large. Following the major themes of the course, here are some general topics for the project:

- *regression* (supervised learning) – use regression methods (e.g. ridge regression, Gaussian processes) to model data or predict from data.
- *classification* (supervised learning) – use classification methods (e.g., SVM, BDR, Logistic Regression) to learn to distinguish between multiple classes given a feature vector.
- *clustering* (unsupervised learning) – use clustering methods (e.g., K-means, EM, Mean-Shift) to discover the natural groups in data.
- *visualization* (unsupervised learning) – use dimensionality reduction methods (e.g., PCA, kernel-PCA, non-linear embedding) to visualize the structure of high-dimensional data.

You can pick any one of these topics and apply them to your own problem/data. Before actually doing the project, you need to write a *project proposal* so that we can make sure the project is doable within the 3-4 weeks. I can also give you some pointers to relevant methods, if necessary.

- *Can my project be my recently submitted or soon-to-be submitted paper?* If you plan to just turn in the results from your paper, then the answer is **no**. The project cannot be be work that you have already done. However, your course project can be based on extending your work. For example, you can try some models introduced in the course on your data/problem.

CS5487 – 作业 3 – 课程项目

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科学系

提案截止日期：第 10 周星期五 展示日
期：第 14 周星期五 报告截止日期：第 14
周星期日

1 课程项目

期末作业是一个由学生自主定义的课程项目。该项目的目标是让你将课程材料用于自己的研究问题，从而获得一些实践经验。如果你想不到项目，那么可以做“默认”项目，即数字分类（见第 2 节）。

1.1 项目主题

该项目的目标是让你将课程材料用于自己的研究问题，从而获得一些实践经验。请记住，做该项目只有大约 4 周时间，因此项目范围不应太大。按照本课程的主要主题，以下是项目的一些通用主题：

- 回归（监督学习）——使用回归方法（例如岭回归、高斯过程）对数据建模或根据数据进行预测。
- 分类（监督学习）——使用分类方法（例如 SVM、BDR、逻辑回归）学习在给定特征向量的情况下区分多个类别。
- 聚类（无监督学习）——使用聚类方法（例如 K 均值、EM、Mean-Shift）发现数据中的自然分组。
- 可视化（无监督学习）——使用降维方法（例如 PCA、kernel-PCA、非线性嵌入）来可视化高维数据的结构。

你可以从这些主题中任选一个，并将其应用到自己的问题/数据中。在真正开始做项目之前，你需要撰写一份项目提案，以便我们确认该项目能在 3-4 周内完成。如有必要，我也可以为你提供一些相关方法的指引。

- 我的项目可以是我最近提交或即将提交的论文吗？如果你打算直接提交论文中的结果，那么答案是不可以。项目不能是你已经完成的工作。不过，你的课程项目可以基于对你工作的扩展。例如，你可以尝试将课程中介绍的一些模型应用到你的数据/问题上。

1.2 Project details

- **Group project** – *all projects* should have a group of 2-3 students. To sign up for a group, go to Canvas ⇒ "People" and then join one of the existing "Project Groups". If you cannot find a group, please use the Discussion board.

- **Project Proposal** – For the first part of the project, you need to write a project proposal. The project proposal should be at most one page with the following contents: 1) an introduction that briefly states the problem; 2) a precise description of what you plan to do – e.g., What types of features do you plan to use? What algorithms do you plan to use? What dataset will you use? How will you evaluate your results? How do you define a good outcome for the project? – The goal of the proposal is to work out, in your head, what your project will be. Once the proposal is done, it is just a matter of implementation!

- **Project Poster Presentation** – Project poster presentations will be at the end of the semester, before the project due date (Week 14). More details will be sent out later. The poster presentation is optional. However, if you want to get an "A" on your project, then you *must* give a presentation. Put it another way, if you don't give a presentation, then you will get at most a "B+" on your project.

- **Project Report** – The project report is essentially the project proposal with all the details filled in. The report should have the following contents: 1) introduction – what is the problem? why is important?; 2) methodology – what algorithms did you use and what are the technical details? what are the advantages and disadvantages?; 3) experimental setup – what data did you use? how did you pre-process the data? which algorithms did you run on the data? what is the metric for evaluation?; 4) experimental results – what were the results? what insight do you get from these results? what are some typical success and failure cases? – The project report should be at least 4 pages. There is no upper page limit, but probably it should not be more than 8 pages long. *For group projects, the project report must state the level of contribution from each project member.*

- **What to hand in –** You need to turn in the following things:

  1. Project proposal (**due Friday, Week 10**).
  2. Project report (**due Sunday, Week 14**).
  3. Presentation poster (**due Friday, Week 14**).
  4. Source code files (**due Sunday, Week 14**).

  Only one group member needs to submit the files on Canvas. *You must submit your course project materials using the Canvas website.* Go to "Assignments" ⇒ "Course Project" ⇒ select the appropriate entry.

- **Third Party Code –** In the course project, you may use 3rd party source code, e.g., libsvm, etc. If you use 3rd party code, you must acknowledge it with an appropriate reference.

- **Grading –** The marks for this project will be distributed as follows:

  – 16.7% – Project proposal.
  – 16.7% – Technical correctness (whether you used the algorithms correctly)

1.2 项目详情

- 小组项目——所有项目都应有一个 2-3 名学生的小组。要报名加入小组，请前往 Canvas ⇒ “People”，然后加入现有的某个“Project Groups”。如果你找不到小组，请使用讨论板。

- 项目提案——对于项目的第一部分，你需要撰写一份项目提案。项目提案最多一页，包含以下内容：1）简要说明问题的引言；2）对你计划做什么的精确描述——例如，你计划使用哪些类型的特征？你计划使用哪些算法？你将使用什么数据集？你将如何评估你的结果？你如何定义项目的良好成果？——提案的目标是在脑海中弄清楚你的项目将会是什么。一旦提案完成，剩下的就只是实现！

- 项目海报展示——项目海报展示将在学期末、项目截止日期之前（第 14 周）进行。更多细节将在之后发送。海报展示是可选的。然而，如果你希望在项目上获得“A”，那么你必须进行展示。换句话说，如果你不进行展示，那么你的项目最多只能获得“B+”。

- 项目报告——项目报告本质上就是填写了所有细节的项目提案。报告应包含以下内容：1）引言——问题是什么？为什么重要？；2）方法——你使用了哪些算法，技术细节是什么？优缺点是什么？；3）实验设置——你使用了什么数据？如何对数据进行预处理？你在数据上运行了哪些算法？评估指标是什么？；4）实验结果——结果如何？从这些结果中得到了什么洞见？有哪些典型的成功和失败案例？——项目报告至少 4 页。页数没有上限，但可能不应超过 8 页。对于小组项目，项目报告必须说明每位项目成员的贡献程度。

- 需要提交的内容——你需要提交以下内容：

  1. 项目提案（第 10 周星期五截止）。
  2. 项目报告（第 14 周星期日截止）。
  3. 展示海报（第 14 周星期五截止）。
  4. 源代码文件（第 14 周星期日截止）。

  只需一名小组成员在 Canvas 上提交文件。你必须使用 Canvas 网站提交课程项目材料。转到“Assignments” ⇒ “Course Project” ⇒ 选择相应条目。

- 第三方代码——在课程项目中，你可以使用第三方源代码，例如 libsvm 等。如果你使用了第三方代码，必须通过适当的引用予以致谢。

- 评分 – 本项目分数分配如下：

  – 16.7% – 项目提案。

  – 16.7% – 技术正确性（是否正确使用了算法）

- 16.7% – Experiments. More points for thoroughness and testing interesting cases (e.g., different parameter settings).
- 16.7% – Analysis of the experiments. More points for insightful observations and analysis.
- 16.7% – Quality of the written report (organized, complete descriptions, etc).
- 16.7% – Project poster presentation.

Note: Here 16.7% means 5/30

– 16.7% – 实验。更全面并测试有趣案例（例如不同参数设置）可获得更多分数。

– 16.7% – 实验分析。有洞见的观察和分析可获得更多分数。

– 16.7% – 书面报告质量（组织有序、描述完整等）。

– 16.7% – 项目海报展示。

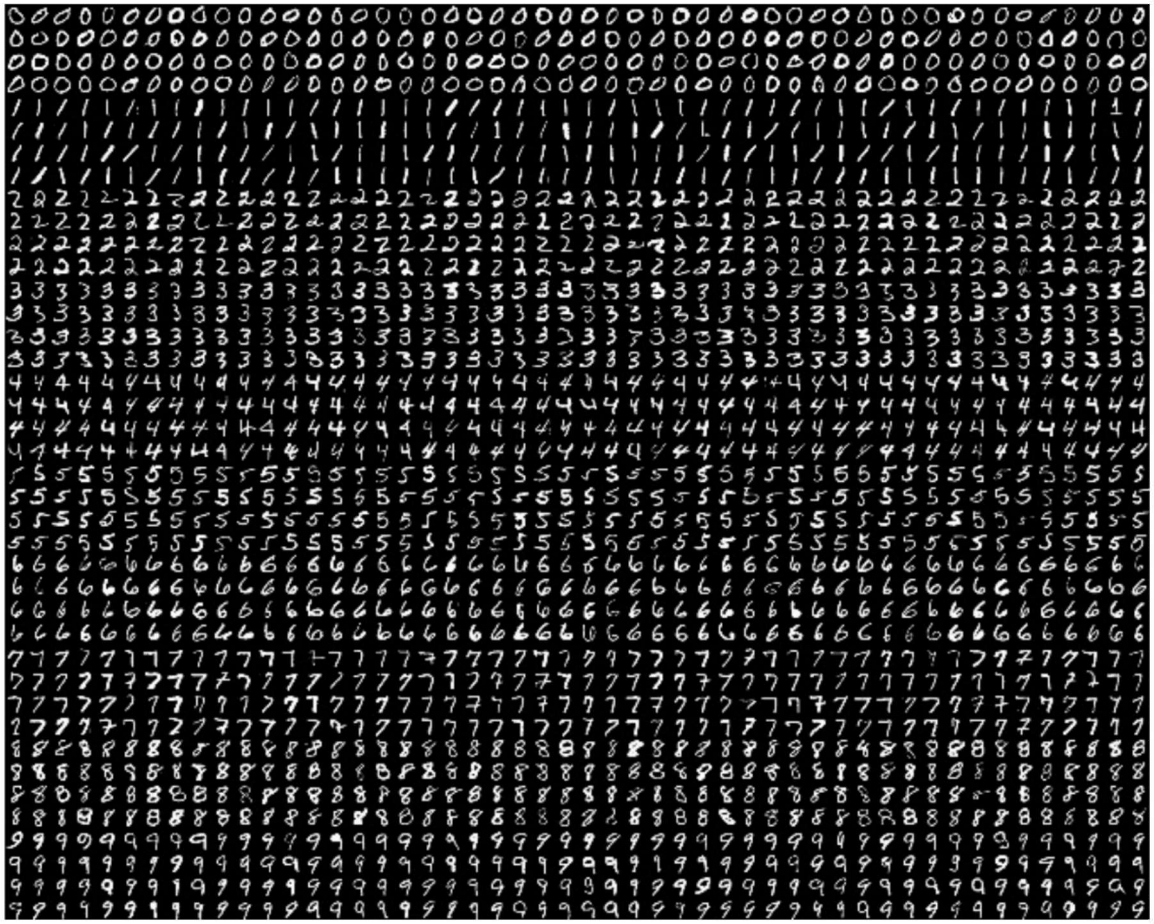
注：此处 16.7% 表示 5/30

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2 Default Course Project – Digit Classification

The default project is handwritten digit classification on a subset of the MNIST digits dataset.

- **Dataset** – The provided dataset is a subset of the MNIST digits. The dataset has 10 classes (digits 0 through 9) with 4000 images (400 images per class). Each feature vector is a vectorized image (784 dimensions), containing grayscale values [0, 255]. The original image dimensions are 28 × 28. Here is an example montage of the digits:



The MATLAB file `digits4000.mat` (or `digits4000_*.txt` for non-MATLAB users) contains the following data:

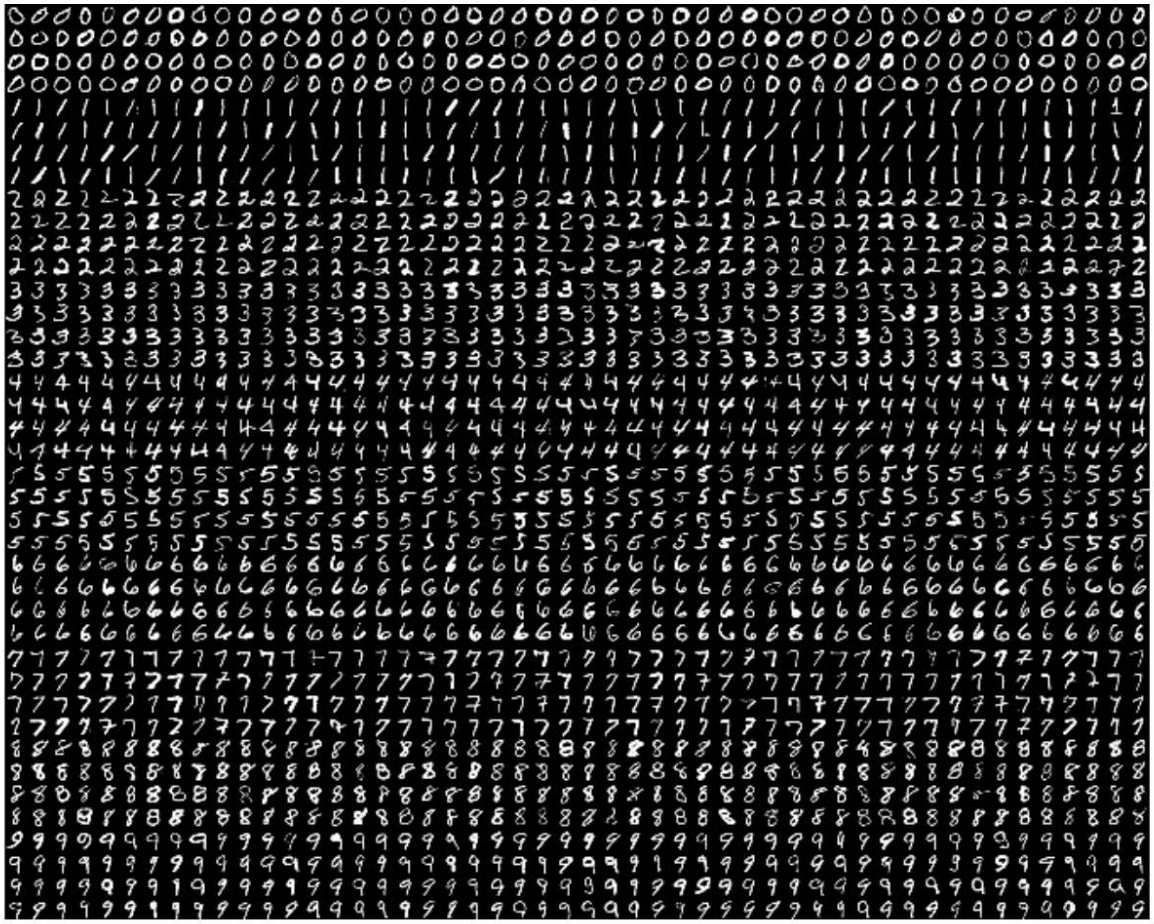
  - `digits_vec` – a 784 × 4000 matrix, where each column is a vectorized image, i.e. the feature vector $x_i \in \mathrm{R}^{784}$.
  - `digits_labels` – a 1 × 4000 matrix with the corresponding labels $y_i \in \{0, \cdots, 9\}$.
  - `trainset` – a 2 × 2000 matrix, where each row is a set of indices to be used for training the classifier.
  - `testset` – a 2 × 2000 matrix, where each row is the corresponding set of indices to be used for testing the classifier.

The image above was generated with the following MATLAB code:

2 默认课程项目——数字分类

默认项目是在 MNIST 数字数据集的一个子集上进行手写数字分类。

- 数据集——所提供的数据集是 MNIST 数字的一个子集。该数据集包含 10 个类别（数字 0 到 9），共有 4000 张图像（每类 400 张图像）。每个特征向量都是一张向量化图像（784 维），包含灰度值 [0, 255]。原始图像尺寸为 28 × 28。以下是数字的示例拼图：



MATLAB 文件 digits4000.mat（非 MATLAB 用户使用 digits4000 *.txt）包含以下数据：

– digits vec – 一个 784 × 4000 矩阵，其中每一列都是一张向量化图像，即特征向量 x∈ R。

– digits labels – 一个 1 × 4000 矩阵，包含对应的标签 y∈ {0, · · · , 9}。

– trainset – 一个 2 × 2000 矩阵，其中每一行都是一组用于训练分类器的索引。

– testset – 一个 2 × 2000 矩阵，其中每一行都是对应的一组用于测试分类器的索引。

上图由以下 MATLAB 代码生成：

```
testX = digits_vec(:,testset(1,:));     % get test data (trial 1)
testXimg = reshape(testX, [28 28 1 2000]); % turn into an image sequence
montage(uint8(testXimg), 'size', [40 50]); % view as a montage
```

- **Methodology** – You can use any technique from the course material, e.g., Bayes classifiers, Fisher's Discriminant, SVMs, logistic regression, perceptron, kernel functions, etc. You may also use other classification techniques not learned in class, but you will need to describe them in detail in your report. Two useful libraries for classification are "libsvm" and "liblinear". You can also pre-process the feature vectors, e.g., using PCA or kPCA to reduce the dimension, or apply other processing techniques (e.g., normalization or some image processing).

  Finally, a common trick for doing multi-class classification using only binary classifiers (e.g. SVMs) is to use a set of *1-vs-all* binary classifiers. Each binary classifier is trained to distinguish one digit (+1) vs. the rest of the digits (-1). In this case, there are 10 binary classifiers total. Given a test example, each binary classifier makes a prediction. Hopefully, only one classifier has a positive prediction, which can then be selected as the class. If not, then the classifier that has the most confidence in its prediction is selected. For example, for SVMs the classifier that places the test example furthest from the margin would be selected. For logistic regression, the selection would be based on the calculated class probability.

- **Evaluation** – The classifiers are evaluated over 2 experiment trials. In each trial, 50% of the data has been set aside for training (and cross-validation of parameters), and the remaining 50% is held out for testing only. The indices of the training set and test sets are given in the `trainset` and `testset` matrices. For a given trial, the same writer does not appear in both the training and test sets.

  For each trial, train a classifier using only the training set data (images and labels). You may also use the training set to select the optimal model parameters using cross-validation. After training the classifier, apply the classifier to the test data (images only) to predict the class. Record the accuracy (number correct predictions / total number) for that trial. *Do not tune the parameters to optimize the test accuracy directly!* You can only tune the parameters using the training set.

  As a baseline, a simple nearest-neighbors classifier with Euclidean distance was used on the test data. The resulting classification accuracy for each experiment trial is:

trial	1	2	mean (std)
1-NN	0.9135	0.9185	**0.9160 (0.0035)**

  In your experiments, which classifier does better? What feature pre-processing helps or hurts the performance? How does the performance vary with parameter values?

- **Bonus Challenge** – In the bonus challenge, I will give you a new test set containing my own handwritten digits, and you will try to classify them using your trained classifiers. Whoever gets the best performance wins a prize!

```
testX = digits_vec(:,testset(1,:)); % 获取测试数据（试次 1）
testXimg = reshape(testX, [28 28 1 2000]); % 转换为图像序列
montage(uint8(testXimg), 'size', [40 50]); % 以蒙太奇形式查看
```

- 方法——你可以使用课程材料中的任何技术，例如贝叶斯分类器、Fisher 判别、SVM、逻辑回归、感知机、核函数等。你也可以使用课堂上未学过的其他分类技术，但需要在报告中详细描述它们。可用于分类的两个有用库是“libsvm”和“liblinear”。你也可以预处理特征向量，例如使用 PCA 或 kPCA 降维，或应用其他处理技术（例如归一化或某些图像处理）。

  最后，仅使用二分类器（例如 SVM）进行多类别分类的一种常见技巧是使用一组一对多（1-vs-all）二分类器。每个二分类器都被训练用于区分一个数字（+1）与其余数字（-1）。在这种情况下，总共有 10 个二分类器。给定一个测试样例，每个二分类器都会做出预测。理想情况下，只有一个分类器给出正预测，然后可以将其选为类别。如果不是这样，则选择对其预测最有信心的分类器。例如，对于 SVM，会选择使测试样例距离间隔最远的分类器。对于逻辑回归，选择将基于计算出的类别概率。

- 评估——分类器在 2 次实验试验中进行评估。在每次试验中，50% 的数据被留作训练（以及用于参数的交叉验证），剩余的 50% 仅留作测试。训练集和测试集的索引在 trainset 和 testset 矩阵中给出。对于给定试验，同一书写者不会同时出现在训练集和测试集中。

  对于每次试验，仅使用训练集数据（图像和标签）训练分类器。你也可以使用训练集通过交叉验证来选择最优模型参数。训练分类器后，将其应用于测试数据（仅图像）以预测类别。记录该试验的准确率（正确预测数 / 总数）。不要直接调整参数以优化测试准确率！你只能使用训练集来调整参数。

  作为基线，在测试数据上使用了采用欧几里得距离的简单最近邻分类器。每次实验试验得到的分类准确率为：

trial	1	2	mean (std)
1-NN	0.9135	0.9185	**0.9160 (0.0035)**

  在你的实验中，哪个分类器表现更好？哪些特征预处理有助于或损害性能？性能如何随参数值变化？

- 附加挑战——在附加挑战中，我会给你一个包含我自己手写数字的新测试集，你将尝试使用训练好的分类器对它们进行分类。表现最佳者将获得奖品！